



Association between prenatal exposure to PM_{2.5} and the increased risk of specified infant mortality in South Korea

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ABSTRACT

Background: Findings from previous studies on the association between exposure to fine particulate matter (PM_{2.5}) and the risk of infant mortality were inconsistent. Thus, two main objectives of our study were to examine the association between exposure to PM_{2.5} and specified infant mortality and to identify critical trimesters.

Methods: We retrospectively created a birth cohort of singleton full-term infants born in South Korea between 2010 and 2015 using national birth and infant mortality data. The specified causes of infant mortality were circulatory and respiratory diseases, perinatal conditions, congenital anomalies, and sudden infant death syndrome. We performed 1:10 propensity score matching for various exposure windows: each trimester, prenatal, and postnatal (up to age 1). Conditional logistic regression was applied to estimate odds ratios (ORs) and 95% confidence intervals (CIs), while accounting for gestational age, birth weight, maternal education level, season of birth, and regions (metropolitan areas/provinces). We also conducted sex-stratified analyses and used different matching ratios for sensitivity analyses.

Results: A total of 2,501,836 births and 761 deaths (0.03%) were identified in the birth cohort. We found an increased risk of infant mortality per 10 µg/m³ increase in PM_{2.5} exposure during the prenatal period (OR: 1.29, 95% CI: 1.07–1.55). Exposure in the 1st and 2nd trimesters was linked to an elevated risk (OR: 1.19, 95% CI: 1.02–1.37; OR: 1.21, 95% CI: 1.04–1.40). However, no association was shown in the third trimester. PM_{2.5} exposure in the 1st and 2nd trimesters was associated with elevated male infant mortality, but did not reach statistical significance in female infants. The use of different matching ratios did not significantly affect the results.

Conclusion: The study findings suggest that exposure to PM_{2.5} could affect infant mortality differently by the timing of exposure and sex, which suggests a relation to fetal development. However, further investigations are warranted.

1. Introduction

Prenatal exposure to environmental hazards could interfere with the fundamental development of a fetus (Selevan et al., 2000). Fine particulate matter (particulate matter with a diameter of < 2.5 µm, PM_{2.5}) is a well-known environmental hazard associated with adverse birth outcomes, including preterm birth, low birth weight, and intrauterine

growth restriction (Bell et al., 2007; Dejmeek et al., 1999; Hyder et al., 2014; Malley et al., 2017) that affect the health of infants. Postnatal exposure to PM_{2.5} is also considered harmful because the pulmonary and immune systems of infants are not yet fully mature (Heinrich and Slama, 2007). The small size of PM_{2.5} allows it to easily penetrate into the alveoli (Xing et al., 2016), enter the bloodstream, and trigger an immune response (Zhang et al., 2018). Therefore, postnatal exposure to

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PM_{2.5} might cause adverse health outcomes in infants.

Previous studies have found a relationship between prenatal and postnatal exposure to PM_{2.5} and an increased risk of infant mortality (Goyal et al., 2019; Heft-Neal et al., 2018; Son et al., 2011; Son et al., 2017a; Woodruff et al., 2008; Woodruff et al., 2006). However, the effect of prenatal exposure to PM_{2.5} on infant mortality is not conclusive because studies have reported contradictory findings (Son et al., 2011; Son et al., 2017a). Similarly, several studies have observed null associations between postnatal exposure to PM_{2.5} and infant mortality (Son et al., 2011; Woodruff et al., 2008; Woodruff et al., 2006), whereas others have found significant associations (Heft-Neal et al., 2018; Son et al., 2017a). Given the inconclusive results in the existing literature, comparisons of different exposure periods during pregnancy and infancy could address knowledge gaps with respect to critical windows of exposure and infant mortality.

Due to the complex nature of infant mortality, finding a plausible relationship between PM_{2.5} exposure and infant mortality can be difficult (Woodruff et al., 2006). Thus, we selected causes of post-neonatal death most attributable to PM_{2.5} exposure based on literature review to examine the association between PM_{2.5} exposure and infant mortality. Deaths of infants less than or equal to 27 days old were excluded because neonatal deaths often occur in hospitals and depend on conditions associated with pregnancy (Woodruff et al., 2008). The early or late effects of PM_{2.5} exposure on morbidity and mortality, such as pregnancy loss and the development of respiratory diseases, were not analyzed either (Jaakkola et al., 2019; Zhang et al., 2019). We then only included singleton full-term births in this study because multiple and preterm births are related to a higher probability of infant mortality (Callaghan et al., 2006; Ko et al., 2018). We selected five specific causes of death for infant mortality, which are respiratory and cardiovascular diseases, congenital anomalies, certain conditions specifically from the perinatal period, and sudden infant death syndrome (SIDS) (Dolk et al., 2010; Pope, 2000; Scheers et al., 2011; Son et al., 2011; Vrijheid et al., 2011; Woodruff et al., 2008).

Advances in medicine have dramatically reduced the infant mortality rate in South Korea to 2.7 per 1000 live births (World Bank Group, 2020). Considering the rarity of this event, a case-control design appeared suitable for examining the association between prenatal and postnatal exposure to PM_{2.5} and infant mortality in South Korea. However, it is important to control fundamental imbalances in covariates, such as disparities in demographic and environmental backgrounds, between cases and controls. Propensity score matching (PSM) can be used to control differences in covariates between cases and controls by generating the same or comparable conditional distributions (Rosenbaum and Rubin, 1983). Finding the optimal matching ratio is complicated because there are advantages and disadvantages to using a small or large number of controls. For instance, choosing a small number of controls can be less beneficial in terms of precise estimation (Dehejia and Wahba, 2002). Thus, comparing results with different matching ratios can increase the validity of study findings.

The aims of this study were to 1) examine the association between exposure to PM_{2.5} and infant mortality and 2) identify critical trimesters. To assess the association, we considered: singleton full-term births only; five selected causes of death, which are respiratory and cardiovascular diseases, congenital anomalies, certain conditions specifically from the perinatal period, and sudden infant death syndrome (SIDS); five windows of exposure across the prenatal and postnatal periods (first, second, and third trimesters, prenatal period, and postnatal period from birth to 1 year of age); and different matching ratios, which are 1:1, 1:2, 1:3, 1:4, 1:5, and 1:10.

2. Materials and methods

2.1. Study population

2.1.1. Data description

We accessed two national population-based datasets, birth and infant mortality data, provided by Statistics Korea for 2010–2015. Birth data contained information from the birth registry, but no further follow-up information on each infant was available. Birth data contained the date of registration, year and month of birth, reported address at birth, place of birth, maternal occupation, maternal education level, maternal age, maternal nationality, paternal occupation, paternal education level, paternal age, paternal nationality, marriage year and month, extramarital birth, sex, multiple birth, birth weight, parity, gestational age, and the number of live births.

Infant mortality data contained information about infants who died before the age of one. Data included the date of death, days of life, reported address at death, place of death, immediate and underlying causes of death, time at delivery, delivery method, multiple birth, birth weight, head circumference at birth, gestational age, Apgar scores at 1 and 5 min, birth order, use of prenatal care, previous pregnancy outcome, maternal age, maternal height, maternal blood pressure before and after birth, maternal weight before and after birth, maternal education level, and maternal nationality.

2.1.2. Creation of birth cohort

We linked birth data and infant mortality data to create a birth cohort of infants born between 2010 and 2015. When linking the two datasets, we identified potential duplicates using variables in common: infant's sex, year of birth, month of birth, administrative division, gestational age (weeks), birth weight (grams), and maternal education level. We used variables in common to detect potential duplicates because personally identifiable information, such as detailed addresses and the names of infants and mothers, was not provided due to the Personal Information Protection Act. In addition, because the birth data did not contain follow-up information on infants, potential cases in birth data had to be manually detected. Detected duplicates were removed from the birth data to avoid inflating the number of cases.

In the birth data, a total of 4477 potential cases were found. Because the number of duplicates was larger than the number of actual cases (3925 cases), it is reasonable to assume that the duplicates included the actual cases. After the duplicates were removed from the birth data, the infant mortality data and birth data were linked.

2.1.3. Inclusion and exclusion criteria

Before linking the two datasets, we excluded 11,545 infants from the birth data and 1180 infants from the infant mortality data because data required for the detection of duplicates were missing. To calculate the birth date of infants in the infant mortality data, the number of prenatal and postnatal days were subtracted from death date because the infant mortality data did not provide the day of birth. We excluded 90 infants from the infant mortality data whose year of birth was 2009. To define the disease categories, the International Classification of Diseases 10th revision (ICD-10) (World Health Organization, 2004) was used. Deaths caused by respiratory and cardiovascular diseases (I00-99, J00-99), congenital anomalies (Q00-99), certain conditions specifically from the perinatal period (P00-96), and SIDS (R95) were selected as the outcomes of interest. This study was approved by the Institutional Review Board at the College of Medicine, Ewha Womans University. Informed content was waived because it was the secondary use of the existing data and both datasets are not individually identifiable.

2.2. Case-control matching

2.2.1. Propensity score matching

We performed PSM using the nearest neighbor algorithm. The

number of controls per case was set to 10 in this study because the cohort contained many more controls than cases (99.97% vs 0.03%). Additionally, the benefit of using more controls is to decrease the variance (Dehejia and Wahba, 2002) and reduce the uncertainty in the estimates. A disadvantage of using a large number of controls, however, is an increased bias (Dehejia and Wahba, 2002) due to the distance (less similarity) between cases and controls. To limit the distance between a case and the farthest control, we applied a caliper to prevent undesirable matches (Caliendo and Kopeinig, 2008). A caliper width of 0.2 was chosen *a priori* (Austin, 2011a).

2.2.2. Matching variables

When selecting variables for PSM, we considered their effects on infant mortality to construct a best model (Brookhart et al., 2006). Thus, the following variables were included for PSM: infant's sex, birth weight (whether equal to or greater than 2500 g), maternal age (years), and average outdoor temperature during each trimester, and in the prenatal and postnatal periods. Sex was included as one of the matching variables to mitigate sex imbalance in the matched sample. We categorized birth weight because low birth weight is a strong predictor of infant mortality (McCormick, 1985). Likewise, maternal age is as an influencing factor in infant mortality (Gage et al., 2008). We also included average outdoor temperature in the PSM separately for each exposure period because mounting evidence suggests a relationship between temperature and an increased risk of infant mortality (Basu et al., 2015; Son et al., 2017b). In addition, average outdoor temperature was used as a proxy for administrative division because infant mortality differs substantially among administrative divisions of South Korea, from 2.6 to 5.6 infant deaths per 1000 births (Korean Statistical Information Service, 2013). The MatchIt package built for R 3.4.4 was used for the PSM (Ho et al., 2018). Table 1 and Tables A.1, A.2, A.3, and A.4 in Appendix show the characteristics of the matched samples, indicating the precision of matching for EPs 1, 2, 3, 4, and 5, respectively. After PSM, the balance of covariates between cases and controls was examined with standardized differences (Austin, 2009).

Table 1
Baseline characteristics of infants for the first trimester (EP 1).

	Birth cohort			Matched sample		
	Controls* (n = 2,501,836)	Cases (n = 761)	p-value**	Controls (n = 7609)	Cases (n = 761)	p-value**
Sex (n, %)			0.17			0.19
Boy	1,279,340 (51.14)	408 (53.61)		3890 (51.12)	408 (53.61)	
Girl	1,222,496 (48.86)	353 (46.39)		3719 (48.88)	353 (46.39)	
Regions (n, %)			0.20			0.17
Provinces	1,399,247 (55.93)	443 (58.21)		4230 (55.59)	443 (58.21)	
Metropolitan areas	1,102,589 (44.07)	318 (41.79)		3379 (44.41)	318 (41.79)	
Maternal education level (n, %)			< 0.0001			< 0.0001
≤ Primary	9400 (0.38)	7 (0.92)		23 (0.30)	7 (0.92)	
≤ Secondary	692,538 (27.68)	310 (40.74)		2147 (28.22)	310 (40.74)	
≥ Post-secondary	1,799,898 (71.94)	444 (58.34)		5439 (71.48)	444 (58.34)	
Season of birth (n, %)			0.22			0.64
Winter (Dec-Feb)	628,255 (25.11)	213 (27.99)		2002 (26.31)	213 (27.99)	
Spring (Mar-May)	640,638 (25.61)	190 (24.97)		1928 (25.34)	190 (24.97)	
Summer (Jun-Aug)	606,796 (24.25)	186 (24.44)		1826 (24.00)	186 (24.44)	
Autumn (Sep-Nov)	626,147 (25.03)	172 (22.60)		1853 (24.35)	172 (22.60)	
Maternal age [years (mean ± SD)]	31.18 ± 4.12	31.51 ± 4.78	0.058	31.33 ± 3.98	31.51 ± 4.78	0.34
Birth weight [kg (mean ± SD)]	3.28 ± 0.38	3.02 ± 0.50	< 0.0001	3.14 ± 0.49	3.02 ± 0.50	< 0.0001
Gestational age [weeks (mean ± SD)]	38.95 ± 1.08	38.64 ± 1.07	< 0.0001	38.81 ± 1.10	38.64 ± 1.07	< 0.0001
Average outdoor temperature [°C (mean ± SD)]			0.85			0.97
EP 1	12.81 ± 9.08	12.88 ± 9.08		12.86 ± 9.02	12.88 ± 9.08	

Abbreviations: SD, Standard deviation; EP 1, Exposure period 1 (first trimester).

* Controls were selected based on propensity score with a case-control ratio of 1:10.

** p-value indicates differences between cases and controls from chi-square or independent t-test.

2.3. Air pollution exposure assessment

2.3.1. Exposure assessment

Because only a few ambient PM_{2.5} monitoring stations were in operation before 2015, daily PM_{2.5} concentrations between 2010 and 2015 were estimated using the Community Multiscale Air Quality (CMAQ) model developed by the United States Environmental Protection Agency. CMAQ, a three-dimensional photochemical air quality model, simulates the spatiotemporal distribution of ground-level particle concentrations using emissions information and meteorological conditions (Byun, 1999). The Kriging procedure was performed to reduce possible biases between the simulated outputs and the available ambient PM_{2.5} monitoring data. The corrected PM_{2.5} outputs were then reconstructed into hourly PM_{2.5} concentrations for each administrative division of South Korea (Han et al., 2018). Detailed information on the verification of the predicted PM_{2.5} data from the CMAQ model can be found elsewhere (Lee et al., 2019).

2.3.2. Spatial coverage

Daily mean PM_{2.5} concentrations were calculated on the basis of the infant's reported address, listed as one of the 16 administrative divisions of South Korea (Fig. 1). The administrative divisions can be grouped into seven metropolitan areas (Seoul, Busan, Daegu, Daejeon, Gwangju, Incheon, and Ulsan) and nine provinces (Gyeonggi-do, Gangwon-do, Chungcheongbuk-do (North Chungcheong; N. Chungcheong), Chungcheongnam-do (South Chungcheong; S. Chungcheong), Jeollabuk-do (North Jeolla; N. Jeolla), Jeollanam-do (South Jeolla; S. Jeolla), Gyeongsangbuk-do (North Gyeongsang; N. Gyeongsang), Gyeongsangnam-do (South Gyeongsang; S. Gyeongsang), and Jeju-do). We linked daily temperature data provided by the National Meteorological Administration of South Korea to the daily PM_{2.5} data using the nearest monitoring station within each administrative division.

2.3.3. Timing of exposure

We considered five exposure periods to find the critical windows of susceptibility: 1) first trimester (from conception to 12 weeks; EP 1), 2)

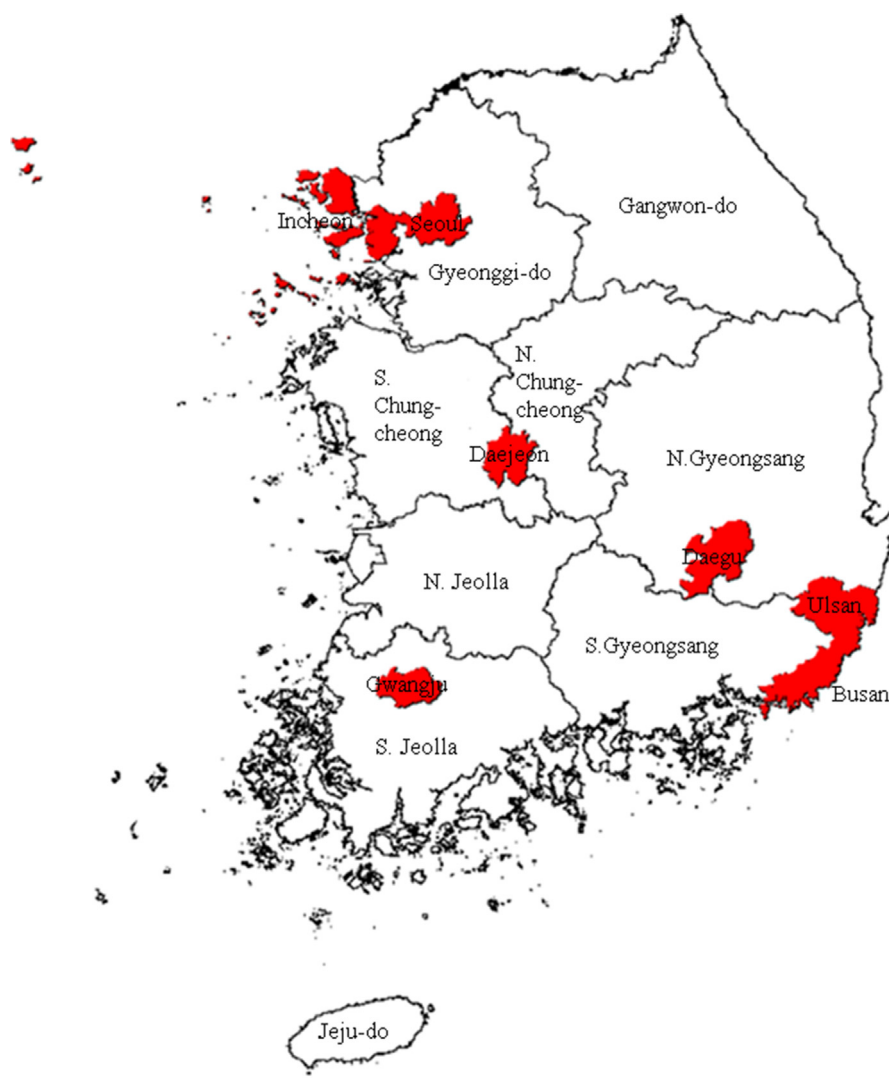


Fig. 1. Spatial coverage of the study, South Korea, 2010–2015: 7 metropolitan areas (in red) and 9 provinces. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

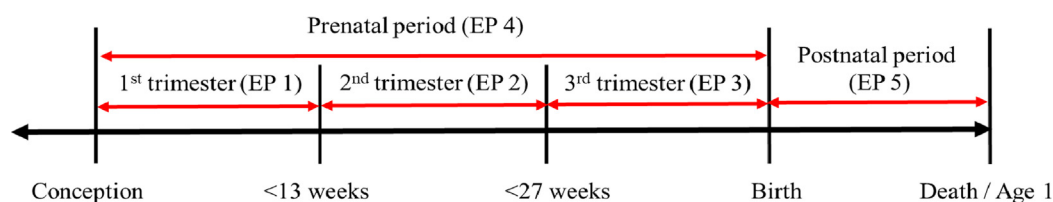


Fig. 2. Schematic representation of the exposure periods: EP 1, Exposure period 1 (first trimester); EP 2, Exposure period 2 (second trimester); EP 3, Exposure period 3 (third trimester); EP 4, Exposure period 4 (prenatal period); EP 5, Exposure period 5 (postnatal period).

second trimester (from 13 weeks to 26 weeks; EP 2), 3) third trimester (from 27 weeks to birth; EP 3), 4) prenatal period (from conception to birth; EP 4), and 5) postnatal period (from birth to death or age 1; EP 5). Fig. 2 displays the scheme of exposure periods. Average $PM_{2.5}$ concentrations and temperatures throughout the exposure periods were assigned to each infant on the basis of their reported administrative division.

For infants in the infant mortality data, the date of conception was retrospectively calculated by subtracting the number of prenatal days from the date of birth. For infants in the birth data, the midpoint of the month of birth was used as the day of birth because the day of birth was unavailable. Prior studies used the same method (Bell et al., 2007; Son et al., 2011).

Because using a different number of days for the five exposure times (EP 1–EP 5) could result in under- or over-estimates, a simulation

scenario was conducted to see the effects of exposure length. We assumed that the whole prenatal period was associated with mortality with no specific critical window. $PM_{2.5}$ was generated at random from a fixed distribution, so that each EP had a comparable mean and standard deviation. That simulation returned an overall under-estimation, the degree of which depended on the length of exposure. Therefore, we compared EP 1–EP 3 (among trimesters of 91–98 days) and the prenatal vs. postnatal periods (EP 4 vs. EP 5 of 280 days and 1 year of age).

2.4. Statistical analysis

Conditional logistic regression (CLR) was applied to estimate the association between a $10 \mu g/m^3$ increase in $PM_{2.5}$ exposure and infant mortality in each EP-specific matched sample (Hosmer et al., 2013). If there are M matched sets and N covariates, the full conditional log-

likelihood is the sum of the log-likelihoods for each matched set. The CLR model can be simplified as follows:

$$\text{logit(OR)} = a_1 + a_2 * I_2 + \dots + a_M * I_M + b_1 * X_1 + b_2 * X_2 + \dots + b_N * X_N + \beta * PM_{2.5}, \quad (1)$$

where the I 's are indicator variables for each matched set, and the X 's are the covariates as potential effect modifiers. The CLR algorithm estimates the b 's and β but not the a 's. Here, the β is of interest to be estimated because it indicates the effect of $PM_{2.5}$ on the odds ratio (OR) in the log scale. OR was adjusted by maternal education level ($\leq$ Primary, $\leq$ Secondary, $\geq$ Post-secondary), season of birth (winter, spring, summer, and autumn), birth weight (kg), gestational age (weeks), and regions (metropolitan areas/provinces).

Sex-stratified analyses were conducted for sex-specific infant mortality. Sensitivity analyses were performed to evaluate the robustness of the 1:10 matched analyses by varying the number of controls per case, 1, 2, 3, 4, to 5. SAS software version 9.4 (SAS Institute, Cary, NC, USA) (SAS Institute, 2014) was used. The threshold for statistical significance was set at 0.05.

3. Results

3.1. Descriptive statistics of the birth cohort and the matched sample for 2010–2015

A total of 2,501,836 births (99.97%) and 761 deaths (0.03%) was identified in the birth cohort. The selected causes were ranked according to the annual number of infant deaths (Fig. 3): the highest was 162 in 2012, and the lowest was 69 in 2015. During the overall study period, more than half of the infants died of congenital anomalies (54.8%, $n = 417$), followed by SIDS (25.36%, $n = 193$), certain conditions from the perinatal period (9.86%, $n = 75$), and respiratory (5.12%, $n = 39$) and circulatory diseases (4.86%, $n = 37$) diseases.

Table 1 presents the characteristics of the cases and controls for the birth cohort and the matched sample for EP 1. The characteristics of the matched samples for EP 2, EP 3, EP 4, and EP 5 can be found in Table A.1, Table A.2, Table A.3, and Table A.4, respectively, in the Appendix. In the birth cohort, more than 50 percent of cases were male (53.61%). Infants living in a province were more likely to die before the age of one (58.21%) than those living in a metropolitan area. Additionally, infants born in winter were at a higher risk of death (27.99%) than those born in other seasons. Even though most of the mothers in both groups (cases and controls) received higher education, a smaller proportion of mothers of cases earned a post-secondary degree (58.34%). Maternal age at birth was slightly advanced for cases (31.51 ± 4.78 years) compared with controls (31.18 ± 4.12 years). Cases were likely to have

lower birth weight (3.02 ± 0.5 kg) and shorter gestational age (38.64 ± 1.07 weeks) than controls (3.28 ± 0.38 kg, 38.95 ± 1.08 weeks). Average outdoor temperature was similar between cases and controls, but a slightly higher average outdoor temperature was shown for cases (12.88 ± 9.08 °C) than for controls (12.81 ± 9.08 °C). The greatest difference in average outdoor temperature between cases and controls was 0.39 °C, which was observed in EP 3 and EP 5.

After 1:10 matching, the matched sample for EP 1 included 8370 infants, with 7609 controls and 761 cases. The number of infants in the matched samples for other EPs, stratified by matching ratios, outcome status (cases/controls), and sex, can be found in Table A.5. Percentage distributions in baseline parameters were similar to those before PSM. The differences in maternal age, birth weight, and gestational age between cases and controls decreased after PSM compared with before PSM. However, even after PSM, averages for those three variables were still higher in the cases. The average outdoor temperature for cases in EP 1 increased after PSM but did not exceed that of controls.

3.2. Assessment of propensity score matching

The standardized differences for the matching variables (infant's sex, birth weight (whether equal to or greater than 2500 g), maternal age (years), and average outdoor temperature during EP 1) are shown in Table 2. The equation for calculating the average standardized difference is described elsewhere (Linden and Samuels, 2013). Even though the standardized difference in the infant's sex did not change, the balance of other covariates was substantially improved by PSM. The most significant improvement was observed in maternal age, with the standardized difference reduced from 0.33 to 0.17 by PSM. For birth weight, the standardized difference decreased from 0.14 to 0. The standardized difference for average outdoor temperature during EP 1 was 0.062 in the birth cohort, which decreased to 0.014 after PSM. PSM also improved imbalances among covariates in the birth cohort for other EPs (Table A.6).

3.3. $PM_{2.5}$ concentrations in different exposure periods

Table 3 presents the average $PM_{2.5}$ concentrations for each EP. Each period showed comparable mean (26.38 to 26.84 $\mu\text{g}/\text{m}^3$) and median values. However, the standard deviation (SD) and interquartile range (IQR) vary for different EPs because they are related to the length of the EP. We found that trimesters had higher SDs and IQRs than the prenatal and postnatal periods. Thus, as the length of the EP gets longer, the SD corresponding to that EP gets smaller. Therefore, we compared ORs of the prenatal and postnatal periods (EP 4 vs. EP 5 of 280 days to 1 year of age) and then subsequently compared the ORs from EP 1–EP 3 (trimesters of 91–98 days) using CLR.

Table 2

Standardized differences in characteristics between cases and controls before and after PSM.

EP 1	Standardized difference before PSM	Standardized difference after PSM
Sex (categorical)		
Boy	0.025	0.025
Girl	−0.025	−0.025
Maternal age (continuous)	0.33	0.17
Birth weight (categorical ^{**})	0.14	0
Average temperature (continuous)	0.062	0.014
Average standardized difference	0.11	0.037

Abbreviations: PSM, Propensity score matching; EP 1, Exposure period 1 (first trimester).

^{**} Below or above 2500 g.

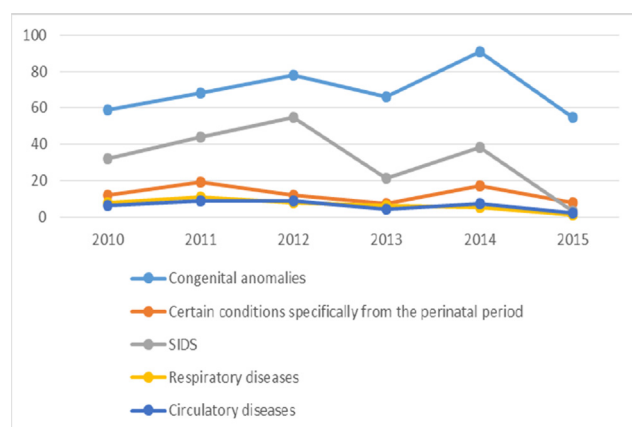


Fig. 3. The annual number of infant deaths in South Korea between 2010 and 2015, by cause.

Table 3
Distribution of average PM_{2.5} concentrations by exposure period (unit: µg/m³).

Exposure period	N	Mean ± SD	Median	IQR
EP 4	2,502,597	26.77 ± 4.64	26.46	5.94
EP 1	2,502,597	26.84 ± 7.08	26.85	10.37
EP 2	2,502,597	26.72 ± 6.96	26.57	10.28
EP 3	2,502,597	26.75 ± 7.15	26.51	10.68
EP 5	2,502,597	26.38 ± 4.07	26.41	5.24

Abbreviations: EP 1, Exposure period 1 (first trimester); EP 2, Exposure period 2 (second trimester); EP 3, Exposure period 3 (third trimester); EP 4, Exposure period 4 (prenatal period); EP 5, Exposure period 5 (postnatal period); SD, Standard deviation; IQR: Interquartile range.

Table 4
Conditional logistic regression estimates for the association between an increase of 10 µg/m³ in PM_{2.5} concentration and infant mortality, South Korea, 2010–2015.

	Overall	Boys	Girls
Exposure period	Adjusted OR* (95% CI)	Adjusted OR* (95% CI)	Adjusted OR* (95% CI)
EP 4	1.29 (1.07, 1.55)	1.34 (1.03, 1.75)	1.13 (0.86, 1.50)
EP 1	1.19 (1.02, 1.37)	1.24 (1.00, 1.53)	1.21 (0.96, 1.52)
EP 2	1.21 (1.04, 1.40)	1.30 (1.05, 1.61)	1.19 (0.96, 1.49)
EP 3	1.07 (0.93, 1.24)	1.03 (0.84, 1.27)	1.06 (0.85, 1.33)
EP 5	0.91 (0.75, 1.11)	1.27 (0.94, 1.72)	0.85 (0.61, 1.17)

Abbreviations: OR, Odds ratio; CI, Confidence interval; EP 1, Exposure period 1 (first trimester); EP 2, Exposure period 2 (second trimester); EP 3, Exposure period 3 (third trimester); EP 4, Exposure period 4 (prenatal period); EP 5, Exposure period 5 (postnatal period).

* Adjusted for gestational age, birth weight, maternal education level, season of birth, and regions (metropolitan areas/provinces).

3.4. Conditional logistic regression results by exposure period and sex

Table 4 provides the ORs and 95% confidence intervals (CIs) for infant mortality per 10 µg/m³ increase in PM_{2.5} concentration. The results of the sex-stratified analyses are also provided in Table 4. A 10 µg/m³ increase in average PM_{2.5} exposure during the prenatal period was significantly associated with infant mortality (OR: 1.29, 95% CI: 1.07–1.55). However, the association was not significant for the postnatal period (OR: 0.91, 95% CI: 0.75–1.11). When examined separately for each trimester, the associations were significant in the first and second trimesters with ORs of 1.19 (95% CI: 1.02–1.37) and 1.21 (95% CI: 1.04–1.40), respectively. No significant relationship was observed in the third trimester (OR: 1.07, 95% CI: 0.93–1.24).

The results of the sex-stratified analyses indicate that prenatal PM_{2.5} exposure produced a significantly higher risk of infant mortality in males (OR: 1.34, 95% CI: 1.03–1.75), while in females the increase did not reach statistical significance. Similar to the findings from the overall sample, a 10 µg/m³ increase in average PM_{2.5} exposure in the postnatal period did not increase the risk of infant mortality among male infants (OR: 1.27, 95% CI: 0.94–1.72). We noted that a 10 µg/m³ increase in average PM_{2.5} exposure in the second trimester was significantly associated with infant mortality among males (OR: 1.30, 95% CI: 1.05–1.61). The OR for the first trimester was 1.24 among male infants (95% CI: 1.00–1.53). A 10 µg/m³ increase in average PM_{2.5} exposure in the third trimester was not significantly related to infant mortality among male infants (OR: 1.03, 95% CI: 0.84–1.27). The associations between exposure to PM_{2.5} and infant mortality were not significant for female infants in any of the exposure periods.

3.5. Sensitivity analysis results

Fig. 4 displays the results of the sensitivity analyses. The plots indicate ORs and 95% CIs for five different matching ratios, stratified by

sex. Even though the matching ratio was altered, statistically significant ORs were again found in EP 1, EP 2, and EP 4 but not EP 3 and EP 5. The results of the sex-stratified analyses with different matching ratios also show that males were at a higher risk of infant mortality than females. Overall, the statistical significance for each EP did not change significantly with changes in the matching ratio.

4. Discussion

In this study, we examined the association between prenatal and postnatal exposure to PM_{2.5} and infant mortality and identified critical trimesters. We created a birth cohort in South Korea between 2010 and 2015 by linking birth data with infant mortality data. We used a case-control design with matching ratios ranged from 1, 2, 3, 4, 5 to 10 and applied CLR to estimate the ORs of infant mortality. Our study findings show a significantly elevated risk of infant mortality with prenatal PM_{2.5} exposure (OR: 1.29, 95% CI: 1.07–1.55) but not with postnatal PM_{2.5} exposure. We also examined narrower time windows within the prenatal period to investigate susceptibility by trimester. We found that exposure to PM_{2.5} in the first and second trimesters significantly affected infant mortality, with ORs of 1.19 (95% CI: 1.02–1.37) and 1.21 (95% CI: 1.04–1.40), respectively. Sex-stratified analyses showed that only males had an increased risk of infant mortality in relation to PM_{2.5} exposure in the prenatal period, particularly during the first and second trimesters. In contrast, females were not significantly affected by PM_{2.5} exposure in any EPs. Sensitivity analyses with different matching ratios were performed to validate the ORs estimated from the 1:10 matching. Even though the number of controls per case was modified, the ORs were comparable with respect to statistical significance.

4.1. Biological mechanisms for pathway, time-varying risk, and sex-difference

The biological mechanisms through which exposure to PM_{2.5} leads to infant mortality is not clearly understood. However, it is widely known that exposure to PM_{2.5} can cause systematic inflammation, oxidative stress, and blood coagulation (Brook et al., 2010). Therefore, if a pregnant woman is exposed to PM_{2.5}, a series of adverse bodily responses could damage the health of the fetus (Kannan et al., 2006). Recent studies have found that PM exposure can also alter placental structure and function (de Fátima Soto et al., 2017; Feng et al., 2016; Valentino et al., 2015), which could hamper fetal growth and development. In fact, PM_{2.5} itself might be able to cross the placental barrier (Wick et al., 2010). Because the immune system of the fetus is in still development (West, 2002), PM_{2.5} that passes directly through the placenta could adversely affect the fetus.

Exposure to a harmful substance during the prenatal period can interrupt the development of multiple systems that are essential for sustaining life. For instance, exposure to PM_{2.5} in the prenatal period could be detrimental to the development of the heart and central nervous system (Moore et al., 2011). In addition, prenatal exposure to PM_{2.5} could impair lung maturity by interfering with cell proliferation and differentiation and lung growth (Schittny, 2017). Given that the disruption of development and the growth of bodily systems before birth is related to adverse health outcomes in life, prenatal exposure to PM_{2.5} can be linked to infant mortality. A previous South Korean study reported estimated hazard ratios (HRs) of 1.53 (95% CI: 1.22–1.90) for all-cause infant mortality and 3.15 (95% CI: 1.26–7.85) for respiratory mortality per IQR increase in prenatal exposure to PM_{2.5} (Son et al., 2011).

The negative effects of PM_{2.5} exposure can differ in magnitude within the prenatal period because the timing of exposure could determine fetal susceptibility (Selevan et al., 2000). Developmentally, exposure to PM_{2.5} in the early prenatal period could be associated with severe defects, whereas exposure to PM_{2.5} in the later prenatal period could be associated with functional defects (Moore et al., 2011). The

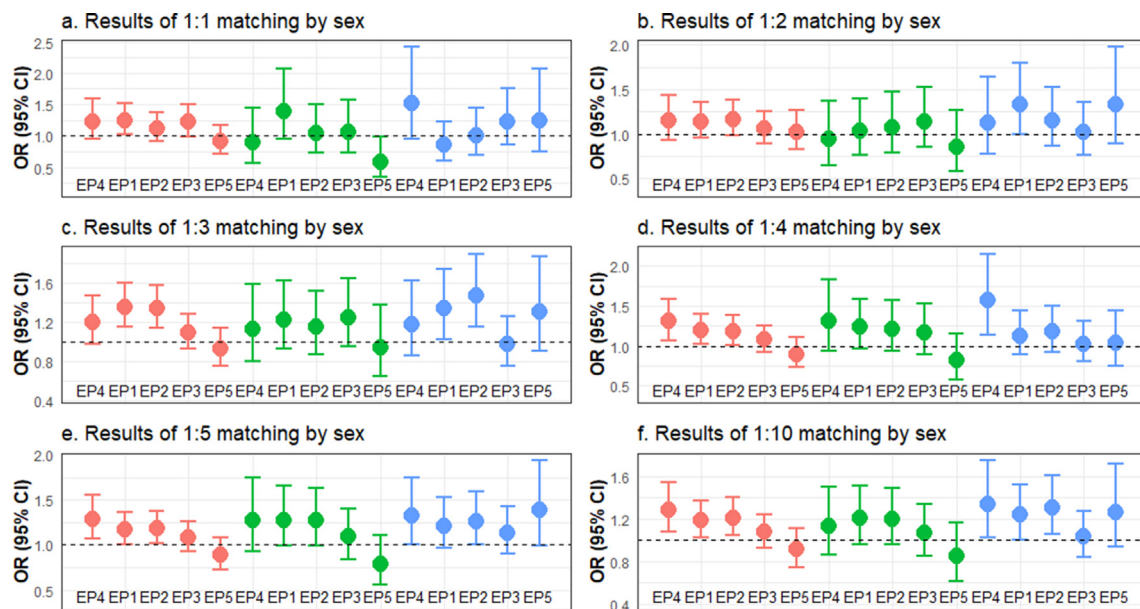


Fig. 4. Odds ratios of overall and sex-specific infant mortality by exposure period for six matching ratios: both sexes in red, females in green, and males in blue. *Adjusted for gestational age, birth weight, maternal education level, season of birth, and regions (metropolitan areas/provinces). Abbreviations: OR, Odds ratio; CI, Confidence interval; EP 1, Exposure period 1 (first trimester); EP 2, Exposure period 2 (second trimester); EP 3, Exposure period 3 (third trimester); EP 4, Exposure period 4 (prenatal period); EP 5, Exposure period 5 (postnatal period). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

results of our study indicate that prenatal exposure to $PM_{2.5}$, especially in the first and second trimesters, could increase the risk of infant mortality. The previous South Korean study showed similar findings: an IQR increase in $PM_{2.5}$ exposure in the first trimester was significantly related to all-cause infant mortality (HR: 1.15, 95% CI: 1.04–1.28) and respiratory infant mortality (HR: 1.58, 95% CI: 1.14–2.19), but no significant associations were observed for the later prenatal period (Son et al., 2011).

However, a study from Massachusetts, United States, found that postnatal exposure to $PM_{2.5}$ could play a significant role in total, respiratory, and SIDS deaths (Son et al., 2017a). Inconsistent findings among studies could arise from different compositions of the $PM_{2.5}$, study design (matched vs non-matched), demographic characteristics of the study population, differences in environmental background, or other factors. The inconsistent results warrant further investigations to better understand the biological mechanisms connecting prenatal exposure to $PM_{2.5}$ and infant mortality.

Findings from our sex-stratified analyses indicate a significant association between prenatal exposure to $PM_{2.5}$ and infant mortality among males. Although it is known that males generally have a higher likelihood of infant mortality than females (Drevenstedt et al., 2008), the delay in fetal lung development in males could explain their increased vulnerability to $PM_{2.5}$ exposure (Torday et al., 1981). Additionally, long-term exposure to $PM_{2.5}$ in the prenatal period can affect the pathway of Ras homolog gene family member A (RhoA) among males (Winckelmans et al., 2017). Because the potential activation mechanism of the RhoA pathway for $PM_{2.5}$ is the increased generation of reactive oxygen species (Lodovici and Bigagli, 2011), males could be more susceptible to cardiovascular disease upon exposure to $PM_{2.5}$ (Winckelmans et al., 2017). Additionally, short-term exposure to $PM_{2.5}$ in the prenatal period can be linked to mitochondrial dysfunction among males (Winckelmans et al., 2017). Because damage to mitochondrial function is related to severe clinical conditions (Wallace et al., 2010), short-term exposure to $PM_{2.5}$ in the prenatal period can also be harmful to the health of male infants. However, the evidence for the observed findings remains insufficient. Therefore, additional epigenetics studies focusing on infants born in South Korea should be conducted to determine the mechanisms of sex-differences in infant

mortality attributable to $PM_{2.5}$ exposure.

4.2. Matching ratio

Changing matching ratios from 1:10 matching to 1:1, 1:2, 1:3, 1:4, and 1:5 did not considerably modify the significance of the ORs in this study. Fullerton et al. (2016) also found that PS matching with different matching ratios (1:1 and 1:3) yielded similar estimated risks (Fullerton et al., 2016). A possible reason is that a balance of covariates between cases and controls was achieved in the matched samples. If the average standardized differences remain relatively stable across the different matching ratios, the effect of the matching ratio can be minimal.

Application of the nearest neighbor matching method with a caliper could be another reason why our ORs did not change significantly with different matching ratios. Given that the proportion of cases was 0.03% in the birth cohort, 1:1 matching could result in a substantial reduction of data in the control reservoir. Increasing the matching ratio would guarantee a sufficient number of eligible infants in the matched sample, which benefits the effect size. Furthermore, a higher matching ratio returns a narrower CI, indicating less uncertainty, and thus has merit over lower matching ratios. Although higher matching ratios provide more precision, they do not necessarily offer more accuracy. Although 1:n matching is known to have higher bias than 1:1 matching, careful use of the nearest neighbor matching method with a caliper could provide more precision with only a small expense in increased bias (Rassen et al., 2012).

4.3. Strengths

We linked national birth data to national infant mortality data and created a birth cohort to ensure that our study sample was representative of the infant population in South Korea. To the extent of our knowledge, this is the first study in South Korea to investigate the association between prenatal and postnatal exposure to $PM_{2.5}$ and infant mortality at the national level. Another advantage of our study is a rigorous methodological approach by which we systematically controlled imbalances between cases and controls and then reduced bias by applying PSM (Austin, 2011b). Additionally, we assessed each infant's

individual exposure history over different EPs to identify critical trimesters. We further conducted sex-stratified analyses to consider the different biological profiles of males and females. Finally, sensitivity analyses with different matching ratios did not significantly alter our main findings, suggesting that our study was not significantly affected by the number of controls.

4.4. Limitations

Possible confounders such as parental smoking, second-hand smoke exposure, and drinking are missing, which is a weakness of this study. Although the prevalence of smoking and alcohol consumption among pregnant women in South Korea has been reported as low as 1.9% and 3.9%, respectively (Lange et al., 2018; Lee et al., 2010), it would have been ideal to account for maternal smoking and alcohol consumption in the analyses. Because parental smoking, drinking, and prenatal secondhand smoke exposure can influence the risk of infant mortality (Ashford et al., 2010; Elliott et al., 2020), the effects of those lifestyle factors, particularly continued use of tobacco and prolonged exposure to secondhand smoke during pregnancy, should be investigated in conjunction with exposure to PM_{2.5} in further studies. Another limitation of our study is that we could retrieve addresses only at the administrative division level. Personal exposure assessment could be used in a further study to strengthen our findings. Furthermore, the exact date of birth would be helpful in assessing personal PM_{2.5} exposure levels. Neonatal deaths were excluded in this study, but they could be separately considered in further, in-depth investigations, as several previous studies have reported associations between PM_{2.5} exposure and neonatal mortality (Lin et al., 2004; Neogi et al., 2015; Proietti et al., 2013). Lastly, it would be interesting to estimate the association between PM_{2.5} exposure and infant mortality for each of the five causes of death considered in this study. However, we did not analyze each cause here, mainly because the number of cases was too small. Table 1 shows that the number of cases from all five causes combined was about 0.03%, so the number from each cause would be much lower than 0.03%, resulting in low statistical power. Therefore, further study is needed to investigate the association between prenatal and postnatal PM_{2.5} exposure and cause-specific infant mortality in South Korea.

5. Conclusions

We found a significantly increased risk of infant mortality in relation to exposure to PM_{2.5} during the prenatal but not the postnatal period. Specifically, early- and mid-pregnancy could be more susceptible to PM_{2.5} exposure than late-pregnancy. We also found males to be at a higher risk of infant mortality than females, suggesting that pregnant women expecting a male infant should be made aware of the elevated risk. The study findings can improve our understanding of the detrimental effects of exposure to PM_{2.5} during the prenatal and postnatal periods. Given that South Korea faces a falling birth rate, it has become a national challenge of how to reduce and prevent infant mortality. The study findings can provide science-based health guidelines for pregnant women to promote infant survival and for policymakers to establish preventive and control measures to minimize infant mortality attributable to PM_{2.5}. For instance, the effectiveness of the current PM_{2.5} standards can be revisited, and an alert system can be introduced as a new risk communication tool especially for pregnant women on a day of high concentration of PM_{2.5}. Furthermore, Asian countries where PM_{2.5} concentrations have increased over the past decades can pay special attention to the adverse effects of PM_{2.5} exposure on infant mortality in relation to South Korea's experience.

This study alone does not provide strong evidence; prospective cohort studies of infant mortality with precise data on day of birth and on exposures of mothers to personal PM_{2.5}, second-hand smoke, alcohol and other risks for infant mortality should be conducted to produce strong evidence.

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CRedit authorship contribution statement

Eun Mi Jung: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization, Writing - review & editing. **Kyoung-Nam Kim:** Writing - review & editing. **Hyeseok Park:** Writing - review & editing. **Hwashin H. Shin:** Formal analysis, Visualization, Writing - review & editing, Supervision (Statistics). **Hae Soon Kim:** Writing - review & editing. **Su Jin Cho:** Writing - review & editing. **Soon Tae Kim:** Investigation, Data curation, Writing - review & editing. **Eun Hee Ha:** Conceptualization, Methodology, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2020.105997>.

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